

SYDNEY TECHNICAL HIGH SCHOOL

MATHEMATICS EXTENSION 1

YEAR 11 YEARLY EXAMINATION

2002

Time allowed: 90 minutes

Instructions:

- Show all necessary working
- Start each question on a new page
- All questions are of equal value
- Marks will not be awarded for careless or badly arranged work
- Non-programmable calculators may be used
- This paper must be handed in with your answer sheets
- Answers must be written in blue or black pen

Name: _____

Class: _____

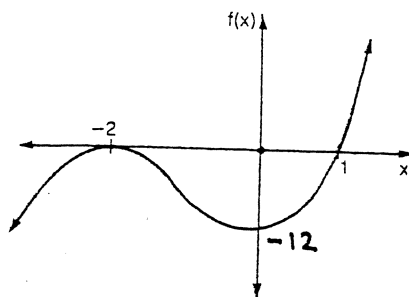
Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	TOTAL

Question 1**Marks**

- a) If $(x + 1)$ is a factor of $P(x) = x^3 - ax + 3$. Find the value of a .

1

b)

2

Write down the equation of the polynomial function (in factored form)

- c) A parabola is symmetrical about the line $y = 2$ it has a focal length 3 units and the equation of the directrix is $x = 1$.

3

- i) How many parabolas satisfy these conditions?
- ii) If the vertex is $(4, 2)$ find the equation of the parabola

- d) Solve $\frac{x+1}{x-1} \leq 0$

2

- e) The roots of the quadratic equation $(k + 2)x^2 - 4x + k^2 = 0$ are reciprocals. Find the value/s of k .

3

Question 2**Marks**

- a) A polynomial of degree 7 is divided by the polynomial $Q(x)$, the remainder is $x^2 + x + 2$. What is the least degree of $Q(x)$. 1
- b) For the quadratic equation $x^2 + (k - 3)x + 2 - k = 0$ 3
- i) Find the value of the discriminant in the terms of k
- ii) Explain why the roots of this quadratic equation are real for all values of k
- c) If $a + b = 1$ and $a^2 + b^2 = 2$ 3
- i) Find the value of ab
- ii) Hence find the value of $a^3 + b^3$
- d) i) Write $x^{-\frac{1}{2}}$ with a positive index 4
- ii) Solve $x^{\frac{1}{2}} + 10x^{-\frac{1}{2}} = 7$

Question 3

Marks

- a) For the function $y = \sqrt{x^2 - 4}$

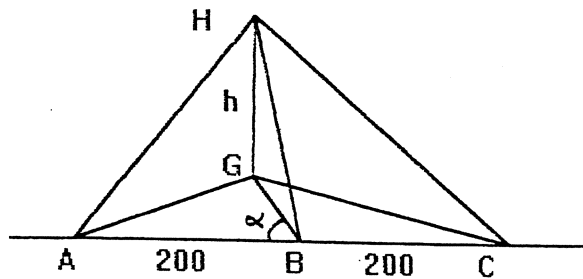
3

Write down

- i) the domain
 - ii) the range
- b) The points $P(12t, 6t^2)$ and $Q(36, 54)$ are points on a parabola
- i) Find the cartesian equation of the parabola
 - ii) If PQ is a focal chord find the value of t

3

c)



A cyclist riding along a straight flat road passes by three stop signs A, B and C spaced 200m apart. From these three signs the respective angles of elevation to the top of a mobile phone tower are $45^\circ, 45^\circ$ and 30° . If 'h' is the height of the tower GH.

5

- i) Show that $CG = \sqrt{3} h$.
- ii) If $\angle GBA = \alpha$. Find two different expressions for $\cos \alpha$ in terms of h .
- iii) Hence find the height of the tower.

Question 4**Marks**

- a) i) Simplify
 $(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$ 3
- ii) The roots of $x^3 - 4x^2 - 8 = 0$ are α , β and γ . Use the result in part (i) to find
The value of $\alpha^2 + \beta^2 + \gamma^2$.
- b) i) Show that $\frac{1 + \cos 2A}{\sin 2A} = \cot A$ 4
- ii) Hence find the exact value of $\cot 15^\circ$
- c) T $(2t, t^2)$ is a point on the parabola $x^2 = 4y$ with focus S. 4
P is the point which divides ST internally in the ratio 1 : 2.
- i) Write down the coordinates of P in terms of t .
- ii) Hence show that as T moves on the parabola $x^2 = 4y$ that
the locus of P is the parabola $9x^2 = 12y - 8$

Question 5

- a) The roots of the equation $x^3 - 6x^2 + 5x + 8 = 0$ are α , β , γ 3
- The roots of the equation $x^3 + ax^2 + bx + 512 = 0$ are $k\alpha$, $k\beta$, $k\gamma$
- i) Find the value of k
- ii) Hence find the value of b .
- b) Consider the points A(-2, 3) B(6, 5) the point P(x, y) moves 4
so that the angle APB = 90°
- i) Write down an expression for the gradient of AP
- ii) Show that the locus of P is a circle
- iii) Find its centre and radius.
- c) i) Expand $\tan(A + B)$ 4
- ii) The roots of $x^2 - 2x - 1 = 0$ are $\tan A$ and $\tan B$. If A and B are acute
find the size of $A + B$.

Question 6**Marks**

- a) i) Derive the equation of the tangent to the parabola $x^2 = 4ay$ at the point $P(2at, at^2)$ 7
- ii) The tangent cuts the y axis at R . Find the co-ordinates of R .
- iii) If S is the focus of the parabola. Find the length of PS .
- iv) Prove that the triangle PSR is isosceles
- v) If $\angle PSR = 120^\circ$. Find the numerical value of t .
-
- b) If $P(x) = 4x^3 + 9x - 4$ 4
- i) Find $P(\alpha + 1)$
- ii) If α is a root of $P(x)$ use part (i) to help show that $P(\alpha + 1) > 0$

Question 1

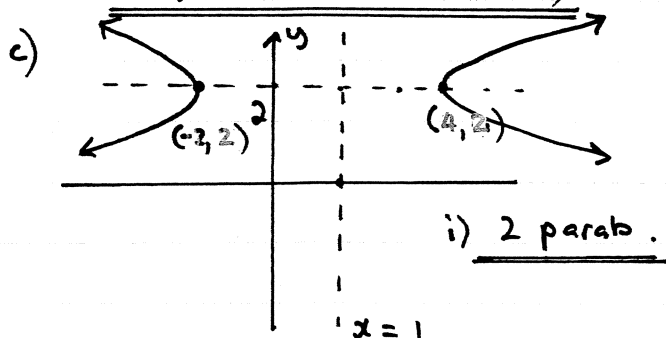
a) $P(-1) = 0 \therefore -1 + a + 3 = 0$

$$\underline{a = -2}$$

b) $P(x) = A(x+2)^2(x-1)$

sub $(0, -12) \therefore A = 3$

$$P(x) = 3(x+2)^2(x-1)$$

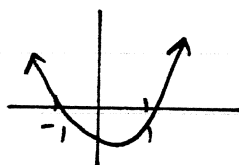


ii) $\underline{(y-2)^2 = 12(x-4)}$

d) $\frac{x+1}{x-1} \leq 0$

$$\frac{(x-1)^2(x+1)}{(x-1)^2} \leq 0 \cdot (x-1)^2$$

$$(x-1)(x+1) \leq 0$$



$$\underline{x: -1 \leq x < 1}$$

e) $\alpha, \frac{1}{\alpha}$ roots

$$\therefore \text{prod} = 1$$

$$\frac{k^2}{k+2} = 1$$

$$k+2$$

$$k^2 = k+2$$

$$k^2 - k - 2 = 0$$

$$(k-2)(k+1) = 0$$

$$\therefore \underline{k = 2, k = -1}$$

$$\therefore \sqrt{x} = 5 \quad \sqrt{x} = 2$$

$$\underline{x = 25} \quad \underline{x = 4}$$

Question 2

a) $Q(x)$ deg 3

$$b) i) \Delta = (k-3)^2 - 4 \cdot 1 \cdot (2-k)$$

$$= k^2 - 6k + 9 - 8 + 4k$$

$$\underline{\underline{\Delta = k^2 - 2k + 1 = (k-1)^2}}$$

ii)

since $\Delta \geq 0$ for all values of k $\therefore$ roots real

c) i) $(a+b)^2 = a^2 + b^2 + 2ab$

$$(a+b)^2 - (a^2 + b^2) = 2ab$$

$$1 - 2 = 2ab$$

$$\therefore \underline{\underline{ab = -\frac{1}{2}}}$$

ii)

$$a^3 + b^3$$

since $(a+b)^3 =$

$$a(a^2 + 2ab + b^2) + b(a^2 + 2ab + b^2)$$

$$a^3 + 3a^2b + 3ab^2 + b^3$$

$$\therefore a^3 + b^3 = (a+b)^3 - 3a^2b - 3ab^2$$

$$= (a+b)^3 - 3ab(a+b)$$

$$= 1 - 3 \times -\frac{1}{2} \cdot 1$$

$$\underline{\underline{= 2\frac{1}{2}}}$$

d) i) $x^{-1/2} = \frac{1}{\sqrt{x}}$

ii) $x^{1/2} + 10x^{-1/2} = 7$

$$\sqrt{x} + \frac{10}{\sqrt{x}} = 7 \quad \text{Let } u = \sqrt{x}$$

$$u + \frac{10}{u} = 7$$

$$u^2 + 10 = 7u$$

$$u^2 - 7u + 10 = 0$$

$$(u-5)(u-2) = 0$$

$$u = 5$$

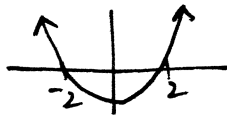
$$u = 2$$

Question 3

a) $y = \sqrt{x^2 - 4}$

$$x^2 - 4 \geq 0$$

$$(x-2)(x+2) \geq 0$$



i) $D: \underline{x \geq 2, x \leq -2}$

(2)

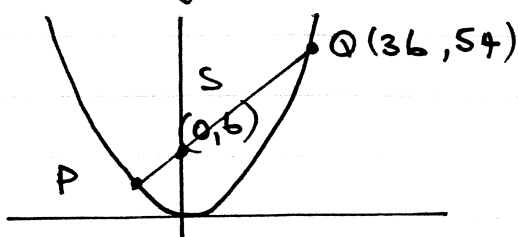
ii) $R: \underline{y \geq 0}$

(1)

b) i) $P(12t, 6t^2)$

$$x = 12t \therefore t = \frac{x}{12} \quad y = 6\left(\frac{x}{12}\right)^2$$

$$\therefore y = \frac{x^2}{24} \quad \text{OR} \quad x^2 = 24y$$



find eqn of SQ

$$m_{SQ} = \frac{4}{3}$$

$$\therefore \text{eqn SQ } y = \frac{4}{3}x + 6$$

sub $P(12t, 6t^2)$

$$6t^2 = \frac{4}{3} \cdot 12t + 6$$

$$6t^2 - 16t - 6 = 0$$

$$3t^2 - 8t - 3 = 0$$

$$(3t+1)(t-3) = 0$$

$$t = 3 \quad t = -\frac{1}{3}$$

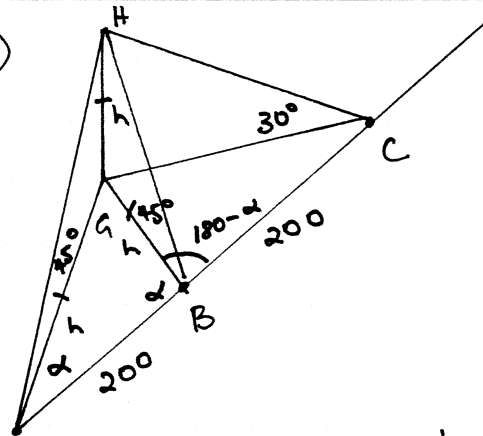
$$t = 3 \Rightarrow Q(36, 54)$$

$$\therefore t = -\frac{1}{3} \Rightarrow P(12t, 6t^2)$$

$$\underline{\underline{P(-4, \frac{2}{3})}}$$

(1)

c)



A i) $\tan 30^\circ = \frac{h}{CG}$

$$\therefore CG = \sqrt{3}h$$

(1)

ii)

In $\triangle ABC \Rightarrow \cos \alpha = \frac{h^2 + 200^2 - h^2}{2 \cdot h \cdot 200}$

$$\cos \alpha = \frac{200^2}{400h}$$

(3)

(1)

$$\cos \alpha = \frac{100}{h}$$

In $\triangle ABC$

$$\Rightarrow \cos(180 - \alpha) = \frac{h^2 + 200^2 - (\sqrt{3}h)^2}{2 \cdot h \cdot 200}$$

$$-\cos \alpha = \frac{20,000 - h^2}{200h} \quad \text{--- E}$$

In $\triangle ACC \Rightarrow \cos \alpha = \frac{h^2 + 400^2 - (\sqrt{3}h)^2}{800h}$

(2)

$$\cos \alpha = \frac{80,000 - h^2}{400h} \quad \text{--- C}$$

any comb of A, B, C.

$$-\frac{100}{h} = \frac{20,000 - h^2}{200h}$$

$$-20000 = 20,000 - h^2$$

$$h^2 = 40,000$$

$$\underline{\underline{h = 200m}}$$

(1)

Question 4

a) i) $(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$

$$= \alpha(\alpha + \beta + \gamma) + \beta(\alpha + \gamma + \beta) + \gamma(\alpha + \beta + \gamma) - 2\alpha\beta - 2\beta\gamma - 2\gamma\alpha$$

$$= \underline{\underline{\alpha^2 + \beta^2 + \gamma^2}}$$

ii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$ (roots)

$a=1 \quad b=-4 \quad c=0 \quad d=-8$

$$= (4)^2 - 2(0)$$

$$= \underline{\underline{16}}$$

b) i) LHS = $\frac{1 + \cos^2 A - \sin^2 A}{2 \sin A \cdot \cos A}$

$$= \frac{2 \cos^2 A}{2 \sin A \cdot \cos A}$$

$$= \frac{\cos A}{\sin A}$$

$$= \cot A$$

$$= \underline{\underline{RHS}}$$

$$= \cot A$$

$$= \cot A$$

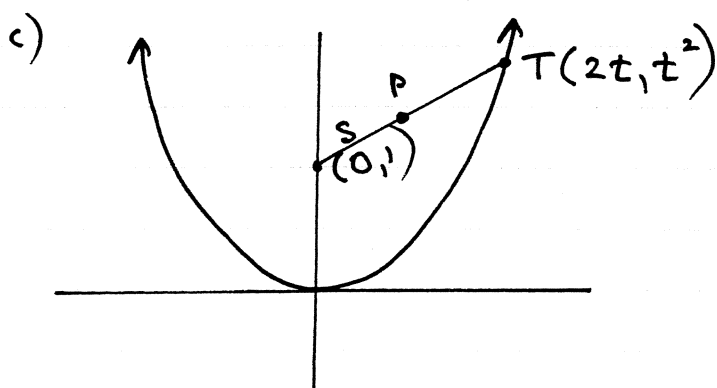
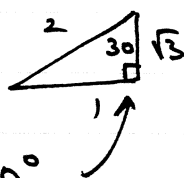
$$= \underline{\underline{RHS}}$$

ii) $\cot 15^\circ = \frac{1 + \cos 30^\circ}{\sin 30^\circ}$

$$= \left(1 + \frac{\sqrt{3}}{2}\right) \div \left(\frac{1}{2}\right)$$

$$= \left(\frac{2 + \sqrt{3}}{2}\right) \times \frac{2}{1}$$

$$= \underline{\underline{2 + \sqrt{3}}}$$



i) $S(0,1) \xrightarrow[1:2]{\text{internally}} T(2t, t^2)$

$$P\left(\frac{0+2t}{3}, \frac{2+t^2}{3}\right)$$

$$\therefore P\left(\frac{2t}{3}, \frac{2+t^2}{3}\right)$$

ii) $T(2t, t^2)$

$$x = 2t \quad y = t^2$$

$$\frac{x}{2} = t \quad y = \left(\frac{x}{2}\right)^2$$

locus of T is $4y = x^2$

$$P\left(\frac{2t}{3}, \frac{2+t^2}{3}\right)$$

$$x = \frac{2t}{3} \quad y = \frac{2+t^2}{3}$$

$$\frac{3x}{2} = t \quad 3y = 2 + \left(\frac{3x}{2}\right)^2$$

$$4 \times 3y = 8 + 9x^2$$

$$\underline{\underline{12y - 8 = 9x^2}}$$

Question 5

a) i) $\alpha + \beta + \gamma = 6$

$$k\alpha + k\beta + k\gamma = -a$$

$$k(\alpha + \beta + \gamma) = -a$$

$$\therefore 6k = -a$$

$$\alpha\beta\gamma = -8$$

$$k^3(\alpha\beta\gamma) = -512$$

$$-8k^3 = -512$$

$$k^3 = 64$$

$$\underline{\underline{k = 4}} \quad \therefore a = -24$$

(4)

$$\alpha\beta + \alpha\gamma + \beta\gamma = 5$$

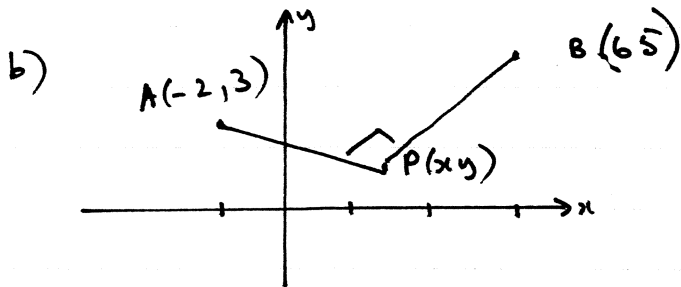
$$k^2\alpha\beta + k^2\alpha\gamma + k^2\beta\gamma = b$$

$$k^2(\alpha\beta + \alpha\gamma + \beta\gamma) = b$$

$$k^2 \cdot 5 = b$$

$$16 \times 5 = b$$

$$\underline{\underline{b = 80}}$$



$$i) m_{AP} = \frac{y-3}{x+2}$$

$$ii) \left(\frac{y-3}{x+2} \right) \cdot \left(\frac{y-5}{x-6} \right) = -1$$

$$y^2 - 8y + 15 = -1(x^2 - 4x - 12) \quad \therefore \text{eqn tang at P } y - at^2 = t(x - 2a)$$

$$y^2 - 8y + 15 = -x^2 + 4x + 12$$

$$x^2 + y^2 - 4x - 8y - 3 = 0$$

$$(x^2 - 4x + 4) + (y^2 - 8y + 16) = 17$$

$$(x-2)^2 + (y-4)^2 = \sqrt{17}^2$$

$$\underline{\underline{\text{centre } (2, 4) \text{ radius } \sqrt{17}}}$$

$$c) i) \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$$

$$ii) \text{Root } \tan A, \tan B$$

$$\text{Sum } \tan A + \tan B = 2$$

$$\tan A \cdot \tan B = -1$$

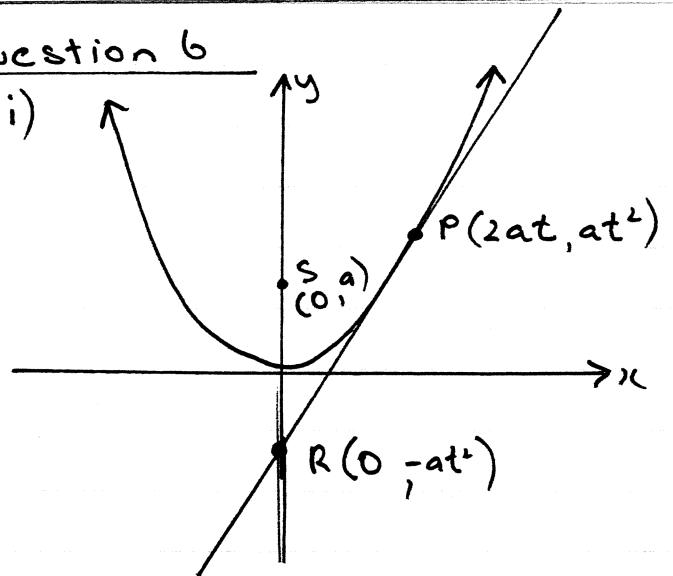
$$\tan(A+B) = \frac{2}{1-1}$$

$$\tan(A+B) = 1$$

$$\underline{\underline{A+B = 45^\circ}}$$

Question 6

a) i)



$$i) x^2 = 4ay$$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$$

$$\text{at } P : m_T = \frac{2at}{2a} = t$$

$$\therefore \text{eqn tang at P } y - at^2 = t(x - 2a)$$

$$\therefore \underline{\underline{y = tx - at^2}}$$

$$ii) \underline{\underline{R(0, -at^2)}}$$

$$iii) PS = \sqrt{(at^2 - a)^2 + (2at - 0)^2}$$

$$= \sqrt{a^2(t^2 - 1)^2 + 4a^2t^2}$$

$$= a\sqrt{t^4 - 2t^2 + 1 + 4t^2}$$

$$= a\sqrt{t^4 + 2t^2 + 1}$$

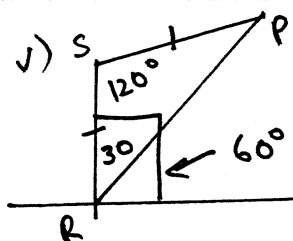
$$= a\sqrt{(t^2 + 1)^2}$$

$$\underline{\underline{PS = a(t^2 + 1)}}$$

$$iv) SR = a + at^2$$

$$= a(1 + t^2)$$

$$SR = PS \therefore \triangle PSR \text{ isosceles}$$



$$\therefore \tan 60^\circ = t$$

(grad of tang)

$$\therefore \underline{\underline{\sqrt{3} = t}}$$

$$b) \quad p(x) = 4x^3 + 9x - 4$$

$$i) \quad p(\alpha+1) = 4(\alpha+1)^3 + 9(\alpha+1) - 4$$

$$ii) \quad \alpha \text{ a root } \therefore$$

$$p(\alpha) = 0$$

$$4\alpha^3 + 9\alpha - 4 = 0$$

$$p(\alpha+1) = 4(\alpha^3 + 3\alpha^2 + 3\alpha + 1) + 9\alpha + 9 - 4$$

$$= 4\alpha^3 + 12\alpha^2 + 12\alpha + 4 + 9\alpha + 9 - 4$$

$$= 4\alpha^3 + 12\alpha^2 + 21\alpha + 9$$

$$= (4\alpha^3 + 9\alpha - 4) + 12\alpha^2 + 12\alpha + 13$$

$$\therefore p(\alpha+1) = \underbrace{12\alpha^2 + 12\alpha + 13}_{\text{+ve def}} \quad \text{since } \alpha > 0 \quad \Delta < 0$$

$$\therefore p(\alpha+1) > 0 \text{ for all } \alpha$$